COMS BC1016

Introduction to Computational Thinking and Data Science

Lecture 13: Statistical Significance

Spring 2025

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November 4, 2026

Updates Starting HW 4

- You should be able to see what the autograder marked incorrect *after* grades on Gradescope are published

- TAs will be manually applying partial credit to autograded multiple choice questions

  - If you got the question correct (or skipped it) the question will be marked as “no correction needed” (+0)

  - Otherwise, TAs will apply points for correctly selecting and correctly not selecting options

Midterm Recap

Last Week's Midterm

- Two versions of the exam
  - Switched some numbers and multiple choice ordering, but same questions except for 2
    - For the different questions, applied similar scoring criteria and strictness
- Version 1 had 3 people score 100 and higher
- Version 0 had a high of ~97 (mean and median were also ~3 points lower than Version 1?)
  - Scores for Version 0 have been curved by +3 points to match Version 1
- Grade distributions are otherwise similar between the two

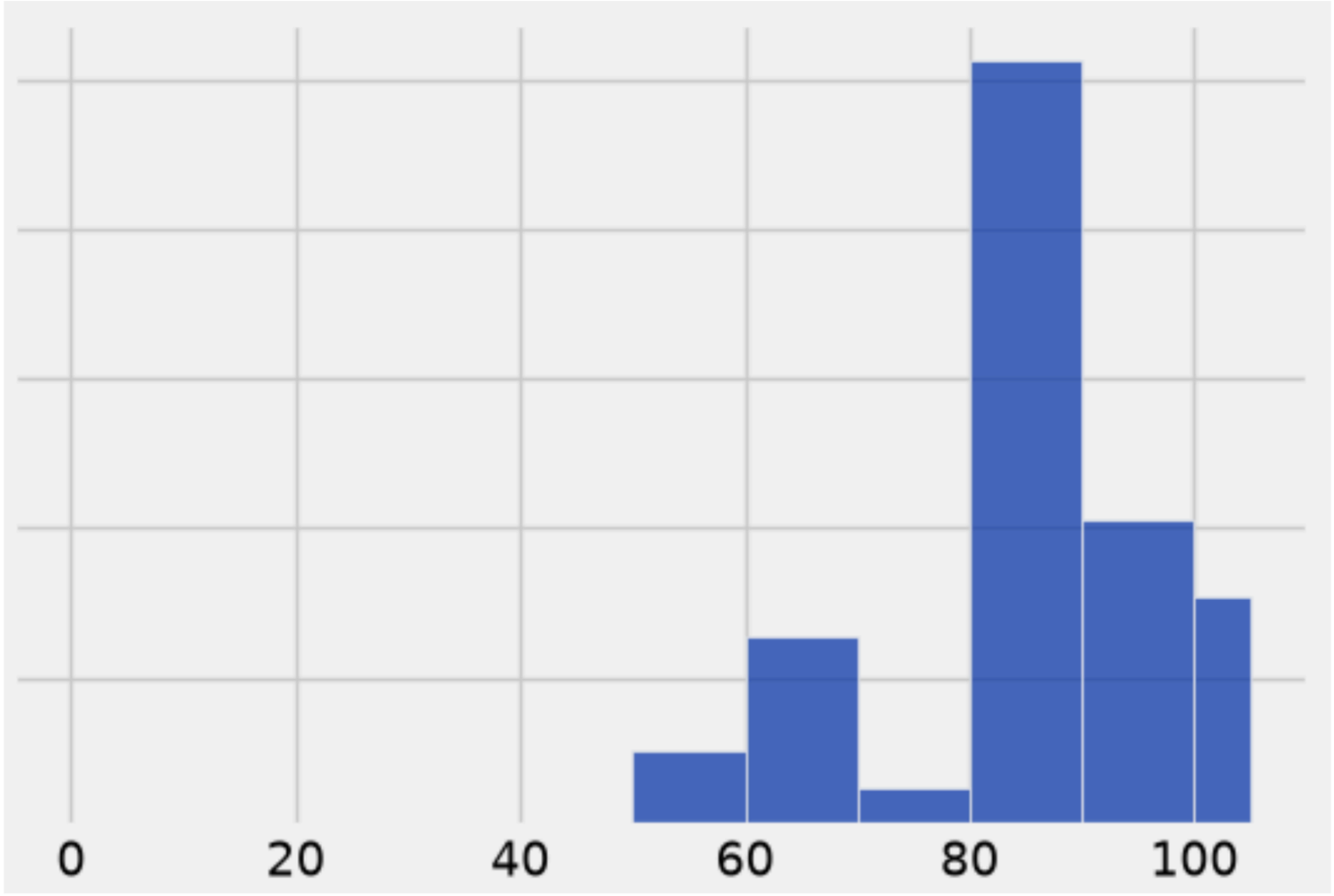
Midterm Stats

Mean: 83.7

Median: 85.5

Bins:

- [50, 70): 7
- [70, 80): 1
- [80, 90): 20
- [90, 100): 8
- [100, 105): 3

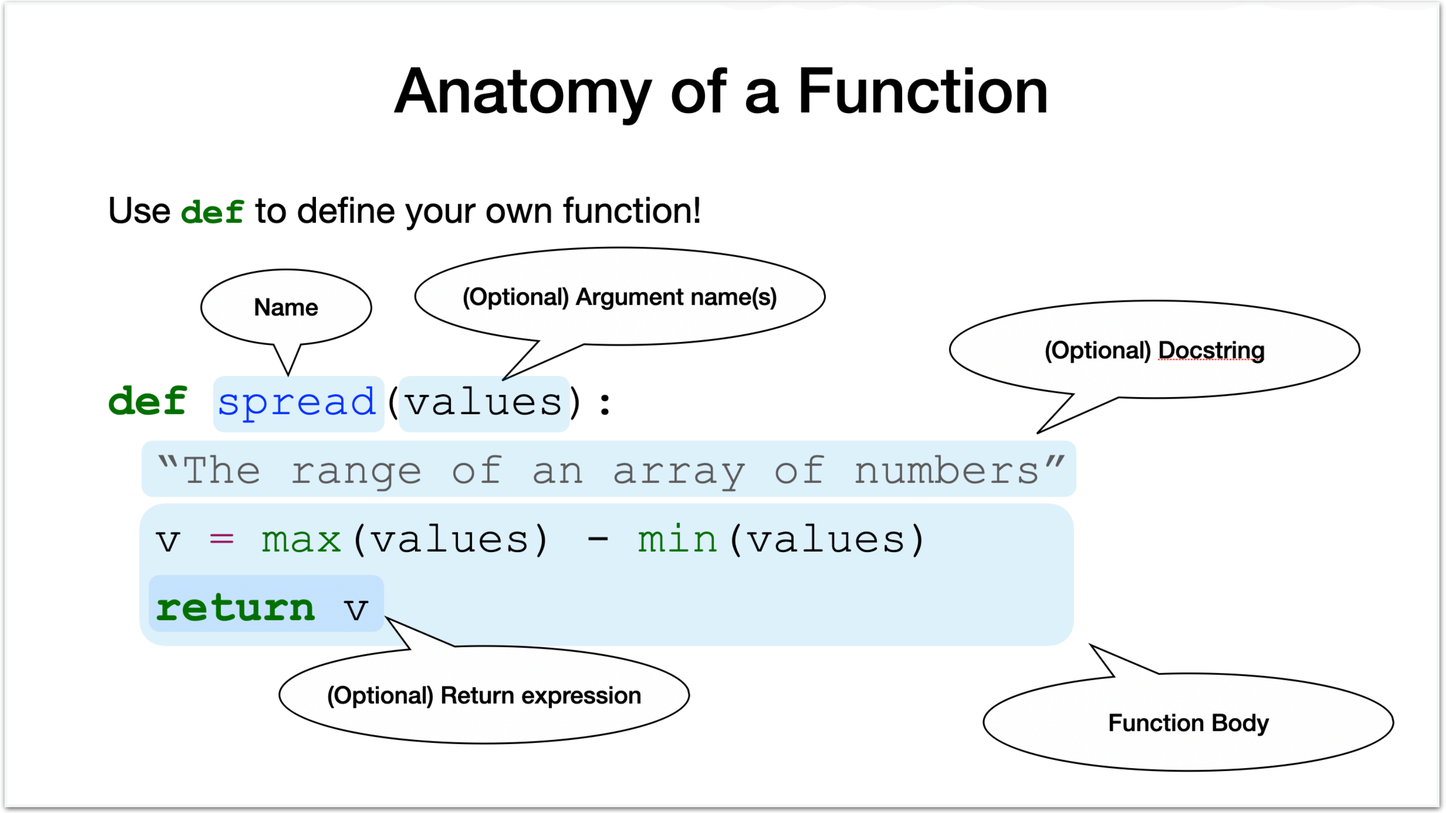


Midterm Sections

- Section 1: Reading Python code + anticipating outputs
- Section 2: Functions and Loops
- Section 3: Visualizations
- Section 4: Probability and Distributions
- Section 5: Terminology + Misc

Midterm Section 2: Functions

- Question 2.1 was alright, but Question 2.2 was the hardest on the exam
- Each decoy answer was selected by at least 25% of the class (up to 50%)



From Lecture 6

Midterm Section 3: Visualizations

- Everyone did very well on attribute types :)
- There was some difficulty with determining visualizations though

Summary: Charts in this Class

Type	Syntax	Description
Line graph	`.plot(x_axis, y_axis)`	Sequential numerical data
Scatter Plot	`.scatter(x_axis, y_axis)`	Non-sequential numerical data
Bar Chart	`.barh(column_label)`	Distribution of a categorical variable *or* relationship between a categorical variable and a numerical values
Histogram	`.hist(column_label, unit, bins)`	Distribution of numerical data

Also from Lecture 6

Midterm Section 3: Groups and Pivots

- You were either asked to cross-classify using group or pivot
- There happened to only be at most 1 entry in each cross-category

Group vs Pivot

Group

- One combo of grouping variables **per row**
- **Any number** of grouping variables
- Aggregate values of **all other columns** in the table
- Missing combos are **absent**

Pivot

- One combo of grouping variables **per entry**
- **Two** grouping variables: columns and rows
- Aggregate values of **values column**
- Missing combos **= 0 (or empty string)**

```
cat_tbl.group(['Sex','Coloring'], np.average)
```

Sex	Coloring	Name average	Age average	Weight average	Owner average
F	tabby		3	7	
F	tortie		6	10	
F	tuxedo		14.5	10	
M	tabby		5	12.25	

```
cat_tbl.pivot('Sex', 'Coloring', 'Weight', np.average)
```

Coloring	F	M
tabby	7	12.25
tortie	10	0
tuxedo	10	0

Midterm Section 4: Probability

- Everyone did well on the first couple probability questions
- Question 4.3 tripped people up. For a die, each roll is independent:
  - Pr[not Yeonji in 2 rolls] = Pr[not Yeonji] * Pr[not Yeonji]

Midterm Section 4: Distributions

Law of Averages / Large Numbers

If a chance experiment is **repeated** many times,

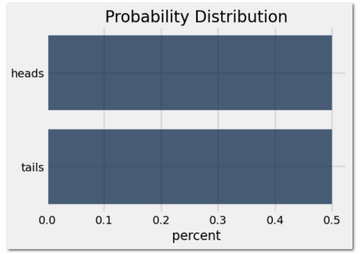
independently and **under the same conditions**,

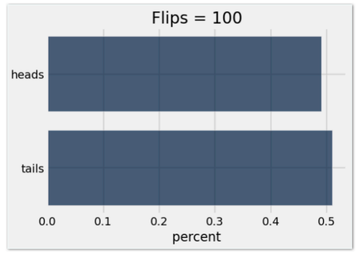
then the **proportion of times that an event occurs**

gets closer to the **true probability of the event**

Distributions

- **Probability Distribution**: All possible values & the probability of each value
  - Example: 50% heads, 50% tails
- **Empirical Distribution**: The observed results (values and outcomes) of an experiment
  - Example: I flipped 100 coins and N of them were heads and rest were tails





From Lecture 10 and 11

Section 5: Terminology

- Sample of Convenience:

(Lecture 10)

If you can't figure out ahead of time

- what's the population
- what's the chance of selection for each group in the population

then it is a **sample of convenience** and not a random sample!

- Parameter vs Statistic:

(Lecture 10 and 11)

Important Terminology

Parameter: Number associated with the population

- Example: average, max, min, mean

Statistic: A number calculated from the sample

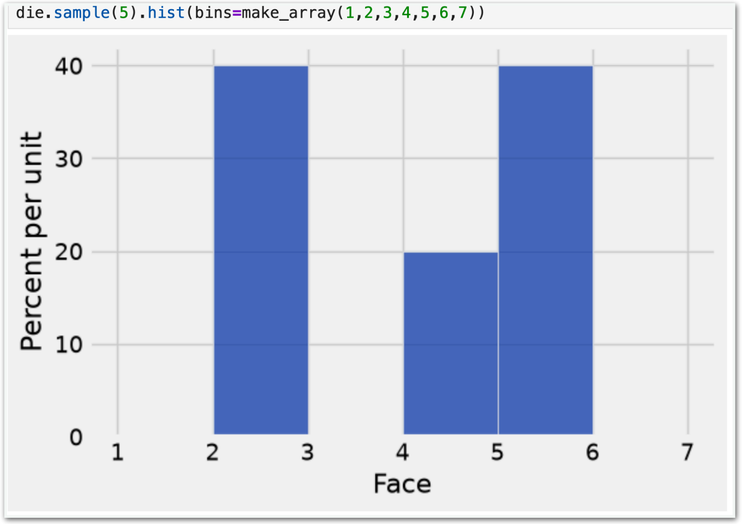
A statistic can be used as an **estimate** of a parameter

- Example: sample mean, sample max, sample min

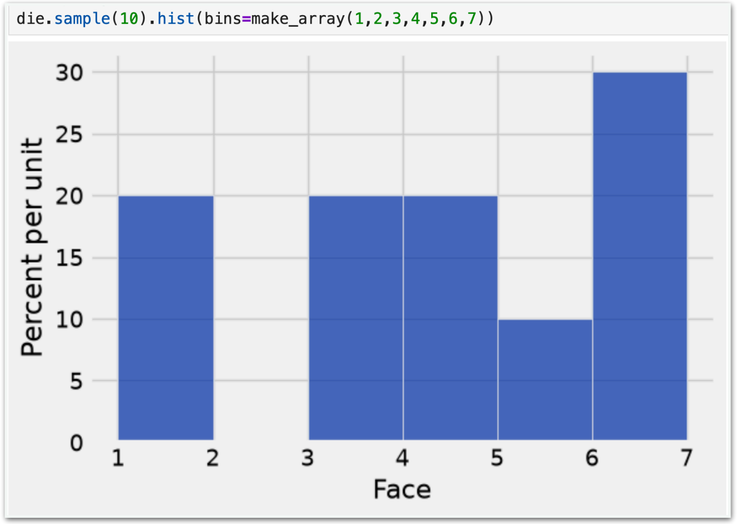
Section 5: Misc

- Why might the list of unique values in a sample be different than the list of unique values in a population?
  - Many possible answers.
  - Example: your sample is smaller than the number of unique values in the population

```
die.sample(5).hist(bins=make_array(1,2,3,4,5,6,7))
```

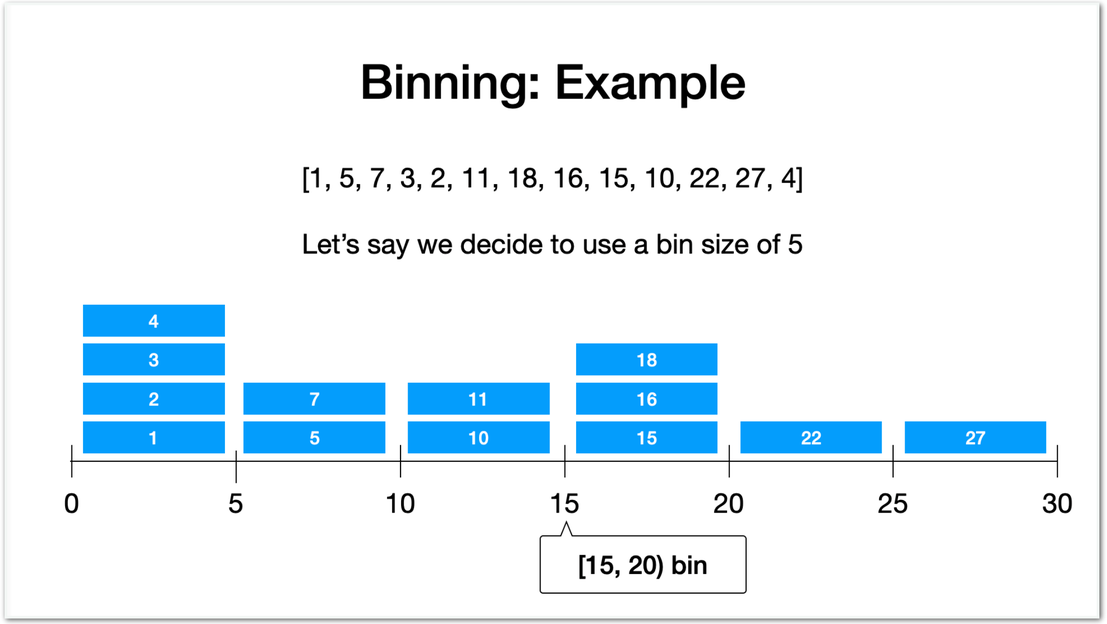



```
die.sample(10).hist(bins=make_array(1,2,3,4,5,6,7))
```



Section 5: Misc

- Is it true that if the [10, 12) bin in my histogram has a very large height it must be that 10, 11, and 12 ALL appear many times in the data?
  - No! 12 is not contained in the bin, and bin specifies a range of values
  - It’s possible one value appears many times and the other doesn’t



Final projects and the rest of the semester

From Lecture 1: Course Outline

- **Exploration**: Identifying patterns and trends using data (e.g., through visualization)
- **Inference**: Drawing reliable conclusions using statistics
- **Prediction**: Making informed guesses about patterns using models

Introduction to Python, working with data — **Week 1 - 5**

Probability, statistics — **Week 6 - 11**

Machine Learning, Regression & Classification — **Week 13 - 15**

Until Thanksgiving: Inference

Today →

11/4	13 - Statistical Significance	Lab 6 - Examining the Therapeutic Touch		
11/6	-			**Final Project Group Declaration** Due
11/9	14 - AB Testing		HW 5 - Testing Hypotheses	
11/11	15 - Confidence Intervals	Lab 7 - A/B Test	HW 6 - Confidence Intervals	
11/16	16 - The Normal Distribution			
11/18	17 - Sample Means	Lab 8 - Normal Distribution and Variability of Sample Means	HW 7 - Sample Sizes and Confidence Intervals	HW 5 Due
11/20	-			**Final Project Proposals** Due
11/23	Special Topics			HW 6 Due
11/25	*Holiday*	*No Lab*		

← Thanksgiving

After Thanksgiving: Prediction

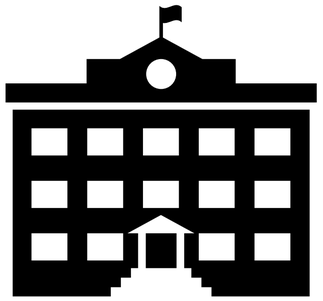
11/20	-			**Final Project Proposals** Due
11/23	Special Topics			HW 6 Due
11/25	*Holiday*	*No Lab*		
11/30	18 - Correlation and Linear Regression		HW 8 - Linear Regression	
12/2	19 - Least Squares, Residuals, and Regression Inference	Lab 9 - Regression		HW 7 Due
12/7	20 - Classification		HW 9 - Classification	**Final Project Progress Reports** Due
12/9	21 - Classification	Lab: Final Project Consultations		HW 8 Due
12/14	22 - Wrap-Up			HW 9 Due
12/18	-			**Final Project Report** Due (No extensions or late days on the final

Thanksgiving

From Lecture 1: Grading Structure

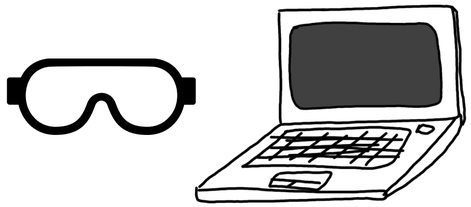
COMS BC1016 (Lecture):

- 30% - Homeworks
- 25% - Midterm
- 35% - Final Project
- 10% - Participation



COMS BC1017 (Lab):

- 50% - Labs
- 50% - Attendance



Final Project

- Work in teams of 2 to explore a real dataset and ask questions that the dataset can help answer
    - We’re currently finalizing some sample data sets for you all to use
- At this point in the semester, we have *not* covered everything needed to complete this project!
    - We want you to get started thinking about who you want to work with and look at the data sets once we’ve released them

Final Project Website

Webpage with details is now up: https://www.eysalee.com/courses/f26/bc1016_final.html

COMS BC1016 Final Project

Fall 2026
Barnard College

The final project for BC1016 provides an opportunity to bring together, apply, and communicate your knowledge of data science and statistics from this course. You will work in groups of 2 to choose one of the provided datasets to analyze and submit a writeup of your analysis and conclusions.

AI and Usage of Outside Resources

As a reminder, the syllabus states:

> For your final project, AI generated text is **not permitted** as part of your written descriptions in your final report. Your report must include your own original writing and reflections. Violations can result in a failing grade for the assignment and/or the course.

Please note this policy includes using generative AI to produce analysis. **You are not allowed to submit AI generated code**.

If you like to do anything more advanced than what we have covered in class, you must include a brief explanation of where you learned these concepts. For instance, if you have taken prior statistics classes and would like to perform a more complicated analysis or if you have experience with (and would like to use) Python libraries we have not covered in class.

Project Milestones and Deadlines

1. **Group Declaration**: Deadline **Friday, Nov 6**
   Please complete the Google Form to indicate your group preference: Final Project Groups. If you do not have a group and would like us to form one

Notable Dates

This week!! →

~2.5 weeks away →

< 5 weeks away →

~ 6.5 weeks away →

1. **Group Declaration**: Deadline **Friday, Nov 6**
   Please complete the Google Form to indicate your group preference: Final Project Groups. If you do not have a group and would like us to f
   for you, there is a space for you to indicate this. Otherwise, you should indicate who you would like to work with. You will be asked to input
   name. You may choose whatever name you like as long as everyone inputs the same name. We will be using this to sort entries.
2. **Project Proposal**: Due **Friday, Nov 20** at 11:59pm
   Each group will select a final project notebook and dataset to work on for the final project and complete the graphs and tables required in t
   Exploratory Data Analysis section. Groups do not need to include sentence explanations of their graphs and tables in the project proposal.
   on your exploratory data analysis, you will state the hypothesis (null and alternative) you are planning to test.
3. **Progress Report**: Due **Monday, Dec 7** at 11:59pm
   For the progress report, groups should complete the exploratory data analysis section and have mostly completed the hypothesis testing s
   Specifically, groups are expected to have a clearly stated null and alternative hypothesis, a clearly stated test statistic, and to have started
   implementing the hypothesis test. The code for your hypothesis does NOT need to work for the progress report. Your group will only be gra
   whether you have started the hypothesis section. Groups should also begin the prediction section. For whatever is not completed in the pre
   section by the time of the progress report, groups should list how they plan on approaching it and what analysis they plan to do. Additionall
   groups should share if they are running into any issues with their analysis that they may need assistance with or have questions about.
4. **Final Project Report**: Due **Friday, Dec 18** at 11:59pm
   Groups will submit the completed reports along with a completed peer review.

Note: We will require all students to complete a peer review to share how work was distributed among team members. Any major discrepancies i
distribution of work will be factored into individual grades on this assignment.

Final Project Grading Breakdown:

Worth 1% of your final project grade!! →

- Group Declaration - 1%
- Project Proposal - 9%
- Progress Report - 25%
- Final Report - 65%

Final Project Groups

- This project is designed to be worked in groups of 2, but you are allowed to work in groups of 1-3
  - Groups of 3 will have slightly more to do to account for the extra teammate
  - Working alone will require the same amount of work as a group of 2
- Fill out a Google Form by Friday, Nov 6
  - Link: https://forms.gle/UmDA2Nx1RS9z1j7F7
  - This is worth 1% of your final project grade



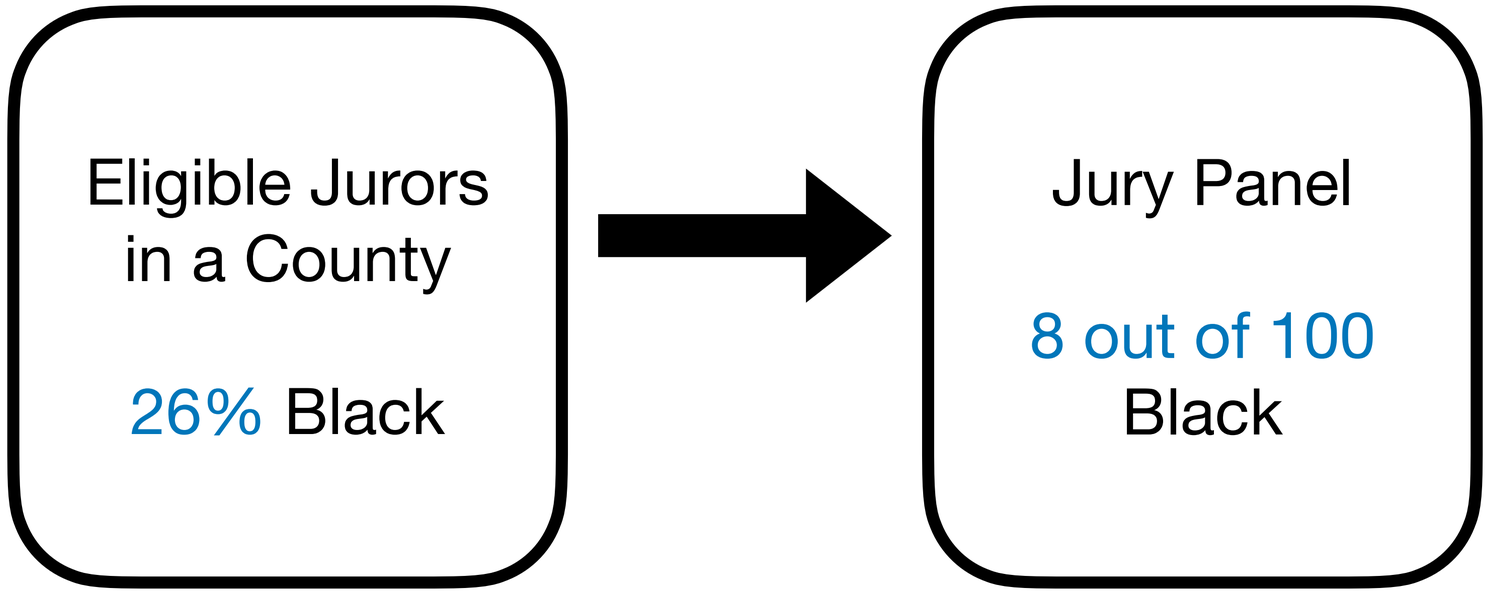
Last Time: Models

Models

A **model** is a set of assumptions about data

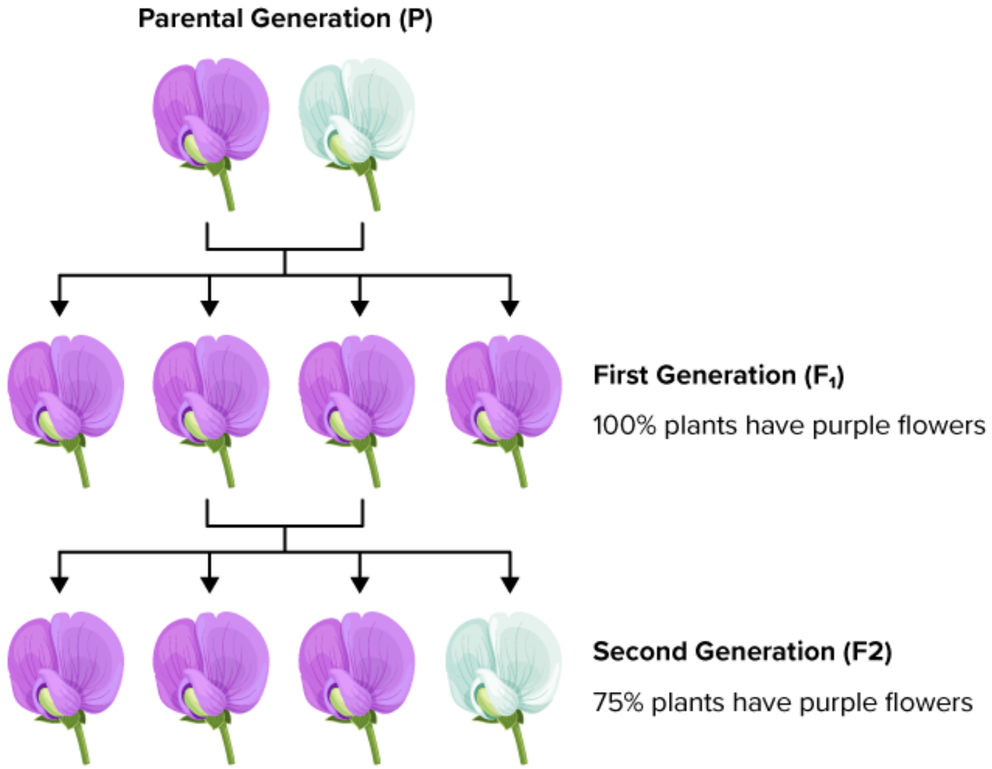
Example: Swain v Alabama

Model: People on the jury panel were sampled randomly from the eligible jurors in the county.



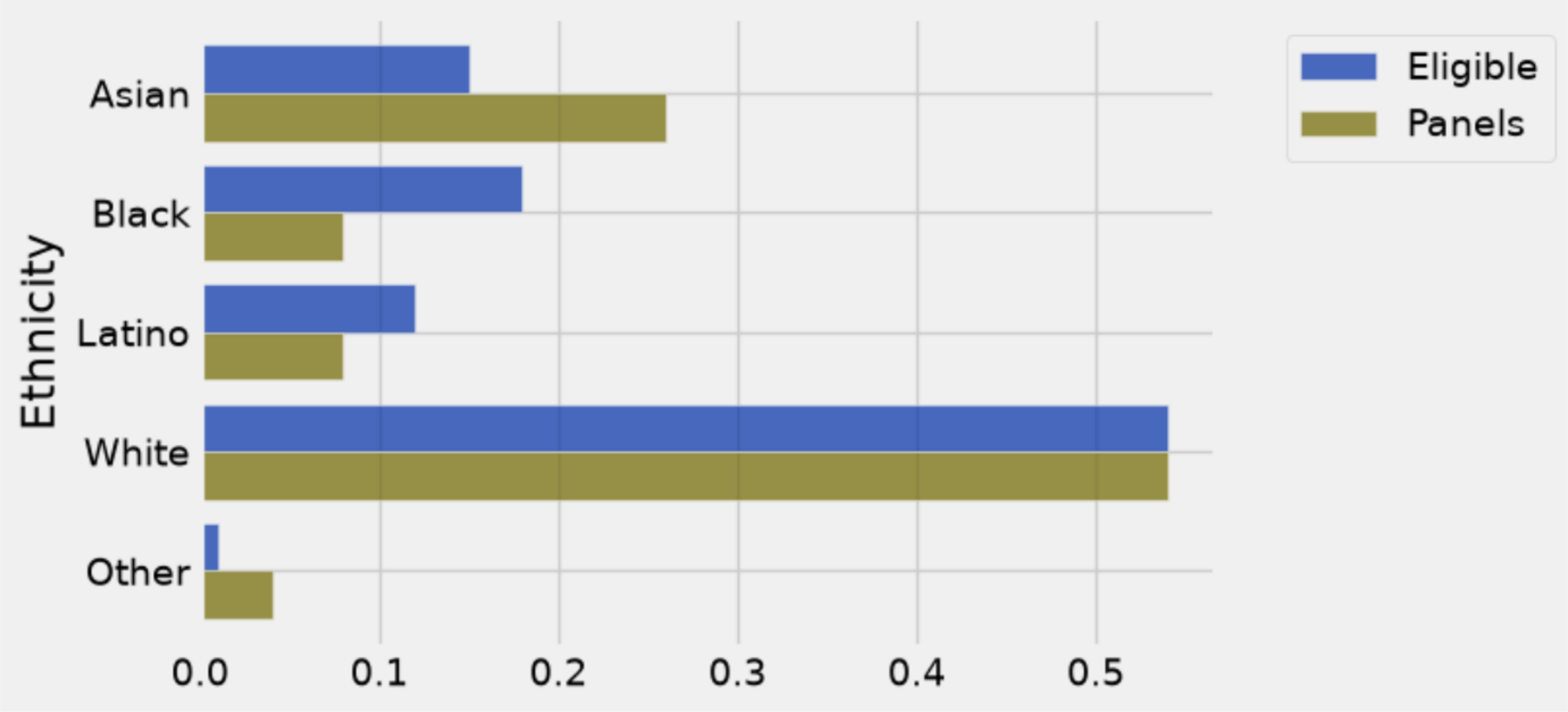
Example: Mendel's Pea Plants

Model: Each second generation pea plant has a 75% chance of having purple flowers.



Example: Alameda County Jury Selection

Model: People on the jury panel were sampled randomly from the eligible jurors in the county.



Assessing Models

Key question: Does the model fit the data?

- How did we assess if the model fits the data?
    - See what the data “should” look like according to the model and compare it to what the data actually looked like
    - If they’re inconsistent, that’s evidence against the model

Steps in Assessing a Model

1. **Choose a statistic** to measure the discrepancy between the model and data

2. **Simulate the statistic** under the assumptions of the model

3. **Compare** the data to the model's predictions

   - Draw a histogram of simulated values of the statistic

   - Compute the observed statistic from the real sample

4. If the observed statistic is far from the histogram, that is evidence against the model

Hypothesis Testing

Testing Hypotheses

- A test chooses between two views of how data were generated
- The views are called hypotheses
- Hypothesis Testing**:** A statistical test in which we choose between two potential views
  - The test picks the hypothesis that is better suited by the observed data

Null and Alternative

The method only works if we can simulate data under one of the hypotheses

- **Null Hypothesis:** A well defined chance model about how the data was generated.
    - We can simulate data under the assumptions of this model (“under the null hypothesis”)
- **Alternative Hypothesis**: A different view about the origin of the data
    - Doesn’t say how or why the chance model isn’t right, just that it isn’t right

Test Statistic

The statistic we choose to simulate to decide between the two hypotheses

Questions before choosing the statistic:

- What values of the statistic will make us lean towards the null hypothesis?
- What values will make us lean towards the alternative?
    - Preferably the answer should just be “high”
    - Try to avoid “both high and low”

Prediction under the Null Hypothesis

- Simulate the test statistic under the null hypothesis
- Draw the histogram of the simulated values
  - This displays the empirical distribution of the statistic under the null hypothesis
- It is a prediction about the statistic made by the null hypothesis
  - It shows all the likely values of the statistic and how likely they are **(if the null hypothesis is true)**
- Remember, probabilities are approximate because we can't generate all possible random samples

Conclusion of the Test

Resolve choice between null and alternative hypotheses

- Compare the observed test statistic and its empirical distribution under the null hypothesis

- If the observed value is **not consistent** with the distribution, then the test favors the alternative (“data is consistent with the alternative”)

Whether a value is consistent with a distribution:

- A visualization may be sufficient

- If not, there are conventions about “consistency” (later in this lecture!)

An Example with Numbers

Scenario: Large Class with 12 Grad Instructors

- A large class is divided into 12 discussion sections
- Each section is led by a graduate student instructor
- After the midterm, students in Section 3 notice the average score in their section is lower than in others

Students wonder: Did Section 3 score lower by chance?

Section	Midterm average
1	15.5938
2	15.125
3	13.6667
4	14.7667
5	17.4545
6	15.0312
7	16.625
8	16.3103
9	14.5667
10	15.2353
11	15.8077
12	15.7333

What can we test?

- Let's think how to formulate this in a way that we can hypothesis test

- Recall: only works if we can simulate the data under one of the hypothesis

Null Hypothesis: A well defined chance model about how the data was generated.

- We can simulate data under the assumptions of this model ("under the null hypothesis")

Alternative Hypothesis: A different view about the origin of the data

- Doesn't say how or why the chance model isn't right, just that it isn't right

Exam Score Hypothesis

- What we want to find out: Did Section 3 score lower by chance?
- Potential Answers:
  - **Null Hypothesis**:
  - **Alternative Hypothesis**:
- **Statistic to measure**:

Exam Score Hypothesis

- What we want to find out: Did Section 3 score lower by chance?

- Potential Answers:

  - **Null Hypothesis**: This average score is in line with if I had chosen the same number of students at random from the whole class

  - **Alternative Hypothesis**: No, the average is too low. Randomness is not the only reason for the low scores

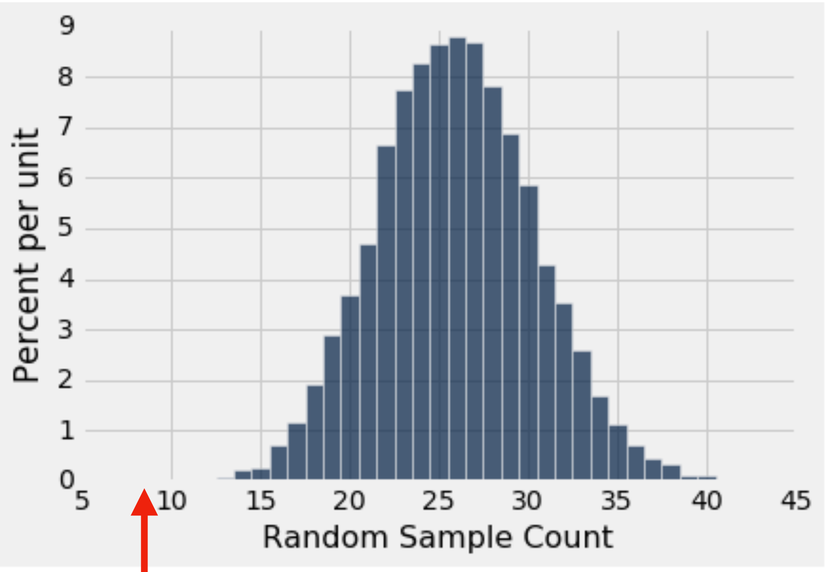
- **Statistic to measure**: Average score per section (27 students)

Demo: Exam Scores

Statistical Significance

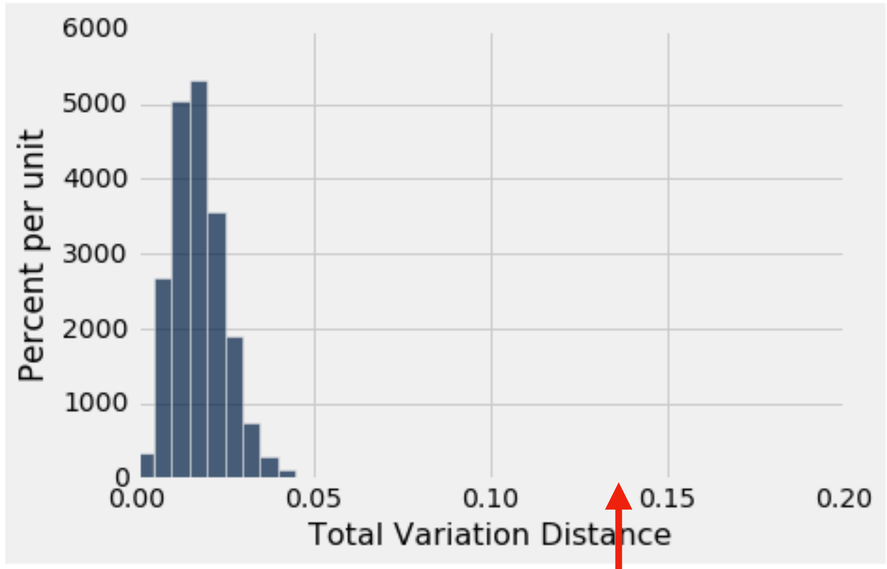
Our Examples So Far

Swain v Alabama



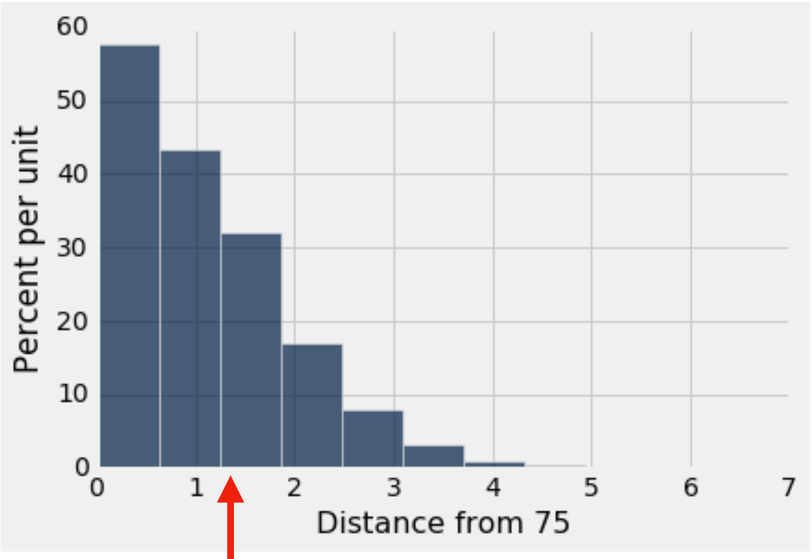
Observed Number (8)

Alameda Jury



Observed TVD (0.14)

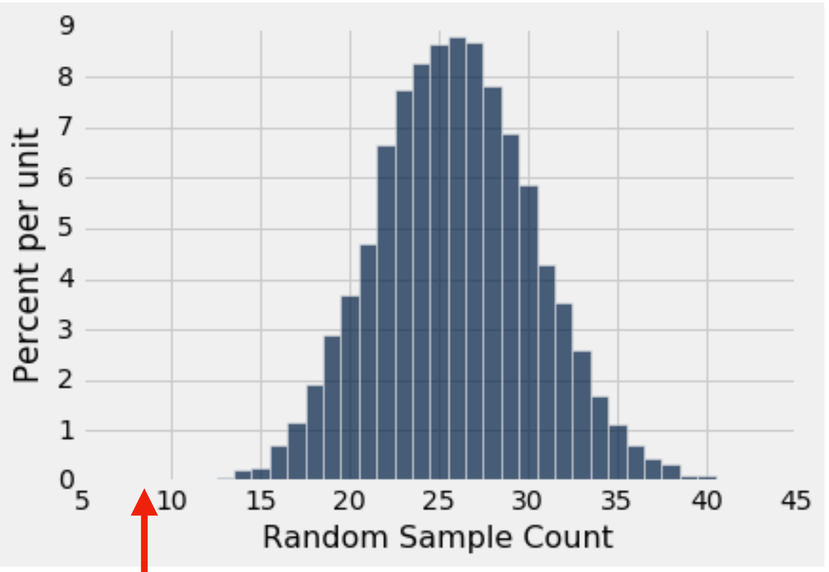
Pea Plants



Observed Distance (1.32)

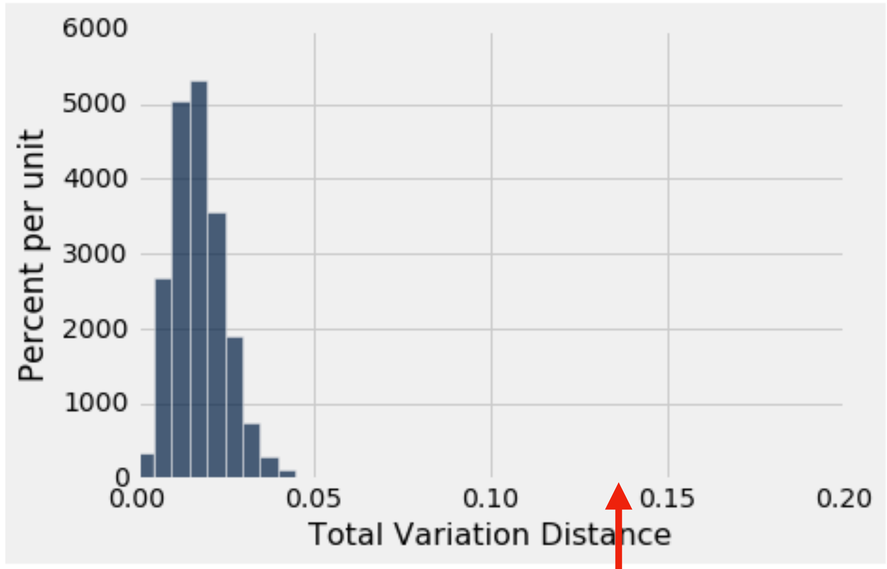
Our Examples So Far

Swain v Alabama



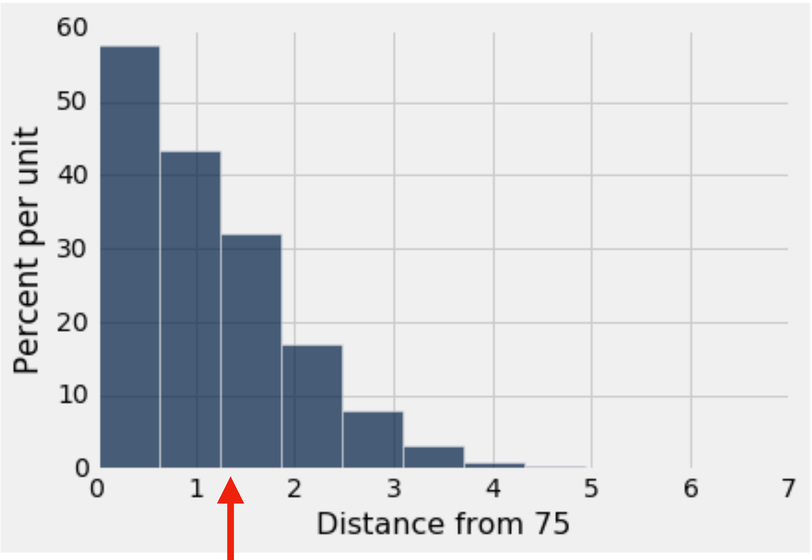
Observed Number (8)

Alameda Jury



Observed TVD (0.14)

Pea Plants

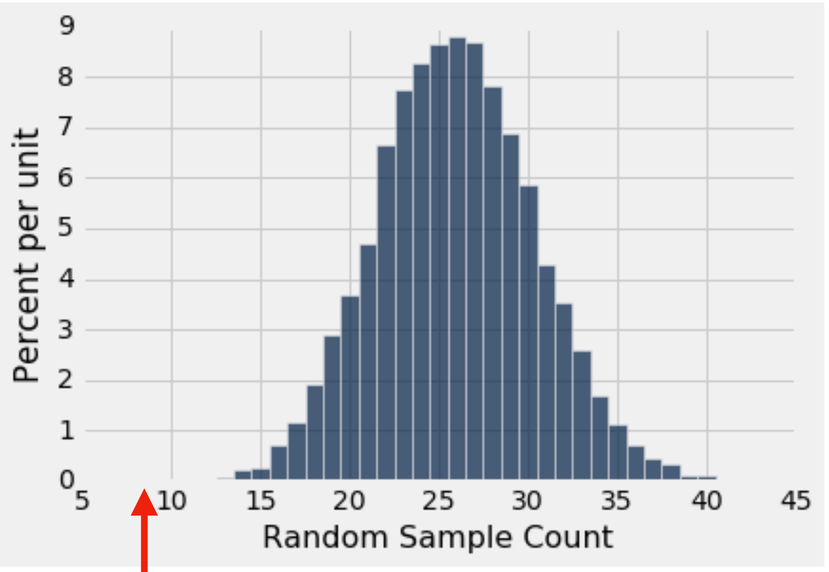


Observed Distance (1.32)

In a histogram, the area of each bar is the percent of individuals in the corresponding bin

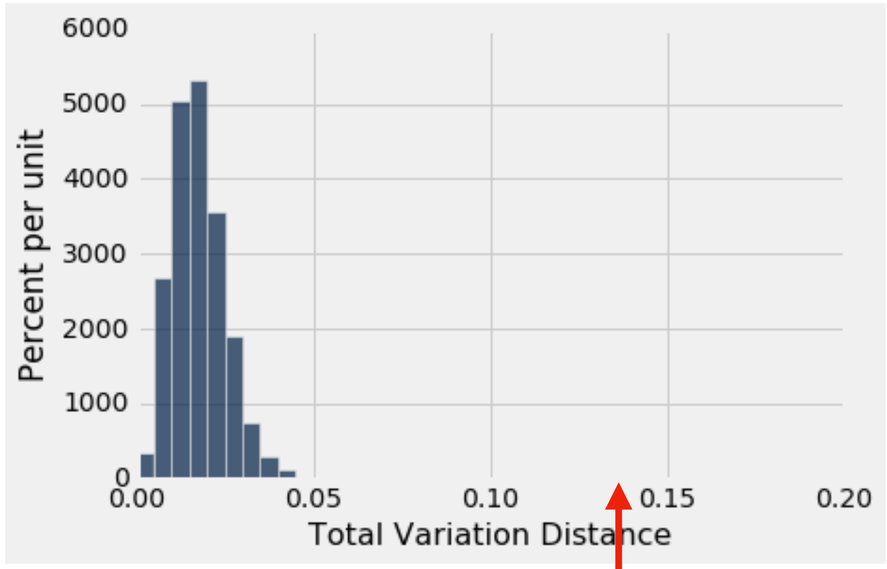
Our Examples So Far: Tail Area

Swain v Alabama



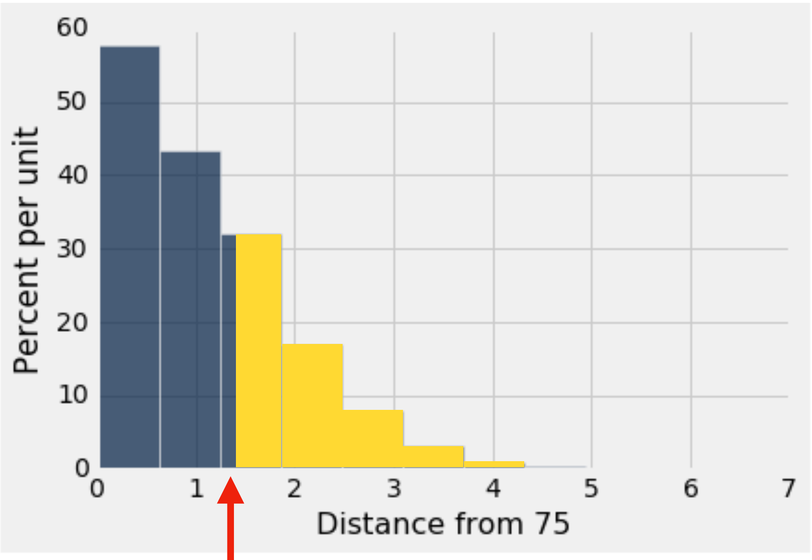
Observed Number (8)

Alameda Jury



Observed TVD (0.14)

Pea Plants



Observed Distance (1.32)

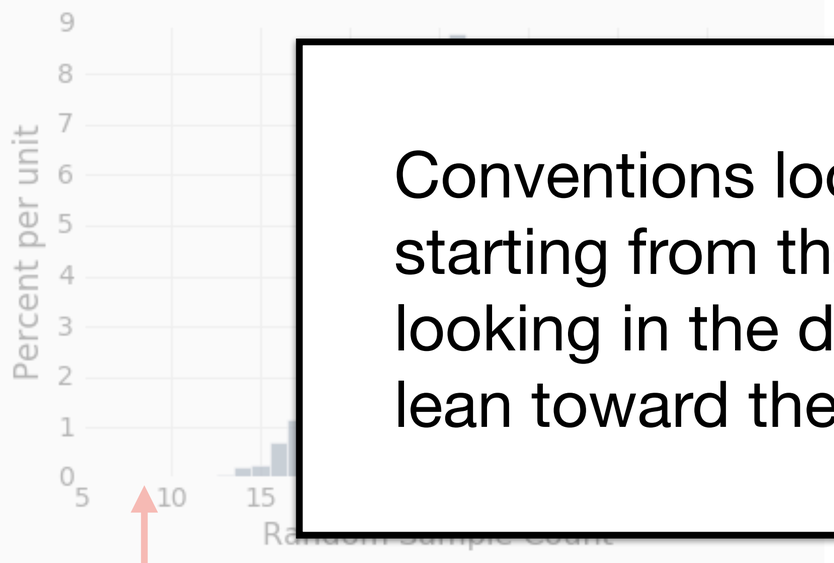
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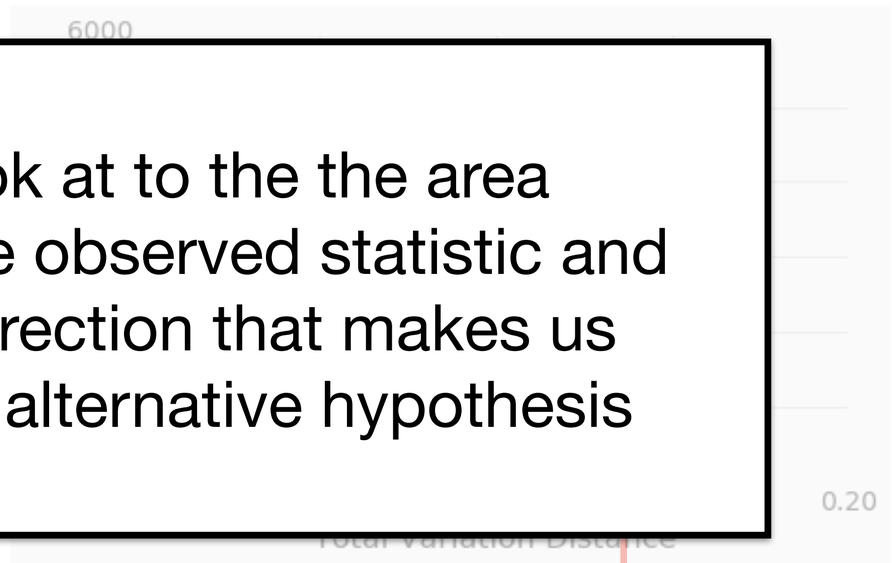
Our Examples So Far: Tail Area

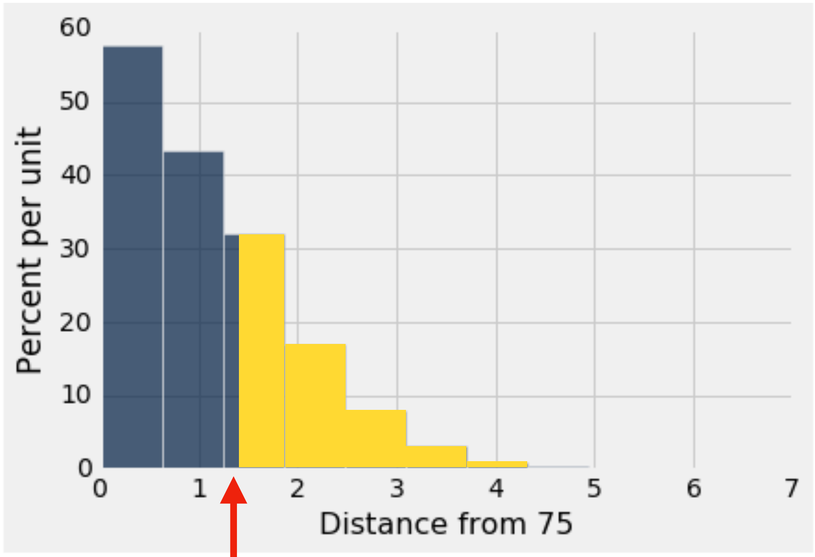
Swain v Alabama

Alameda Jury

Pea Plants







Conventions look at to the the area starting from the observed statistic and looking in the direction that makes us lean toward the alternative hypothesis

Observed Number (8)

Observed TVD (0.14)

Observed Distance (1.32)

In a histogram, the area of each bar is the percent of individuals in the corresponding bin

Conventions about Inconsistency

- **“Inconsistent with the null”**: The test statistic is in the tail of the empirical distribution under the null hypothesis
- **“In the tail”, first convention**:
  - Area in the tail is less than 5%
  - The result is “statistically significant”
- **“In the tail”, second convention**:
  - Area in the tail is less than 1%
  - The result is “highly statistically significant”

Definition of the *P*-value

Formal name: observed significance level

The ***P*-Value** is the chance

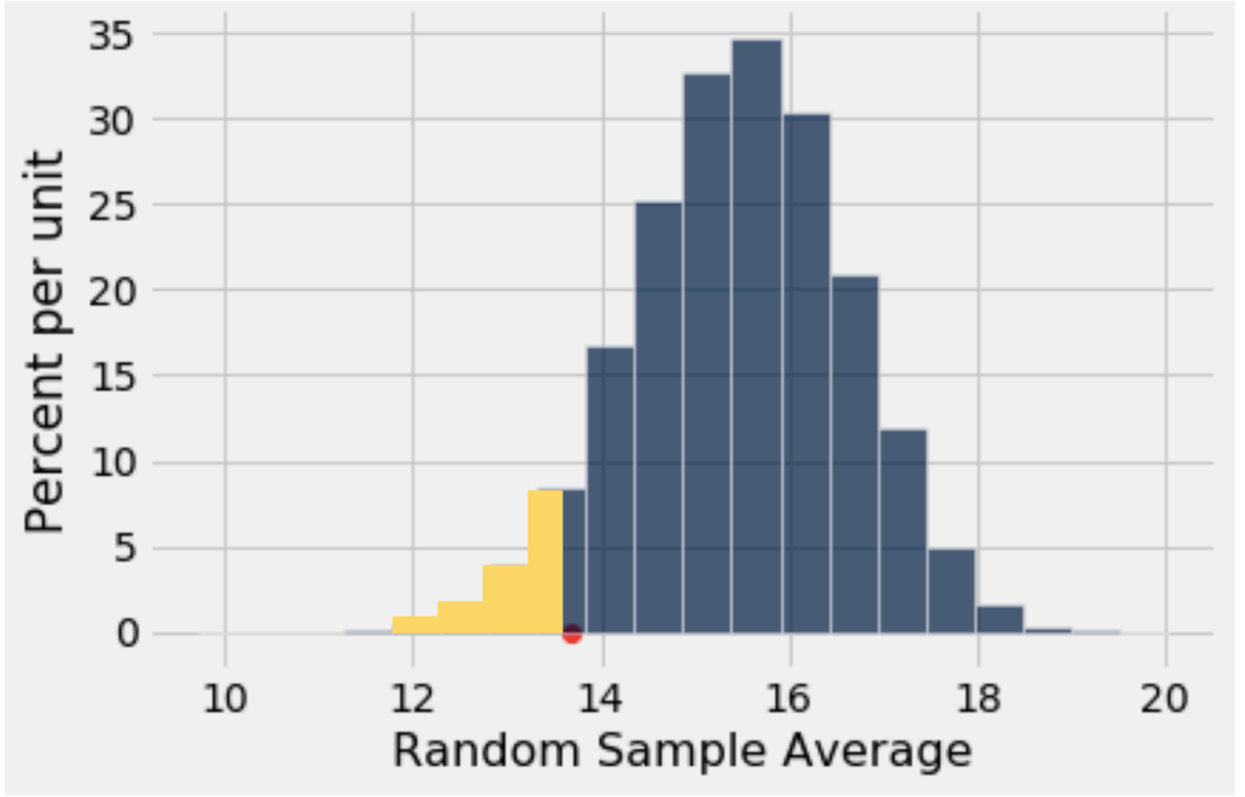
- under the null hypothesis
- that the test statistic
- is equal to the value observed in the data
- or is even further in the direction that supports the alternative

Exam Notebook Demo (continued)

The P-Value as an Area

Empirical distribution of the test statistic under the null hypothesis

The red dot is the observed statistic



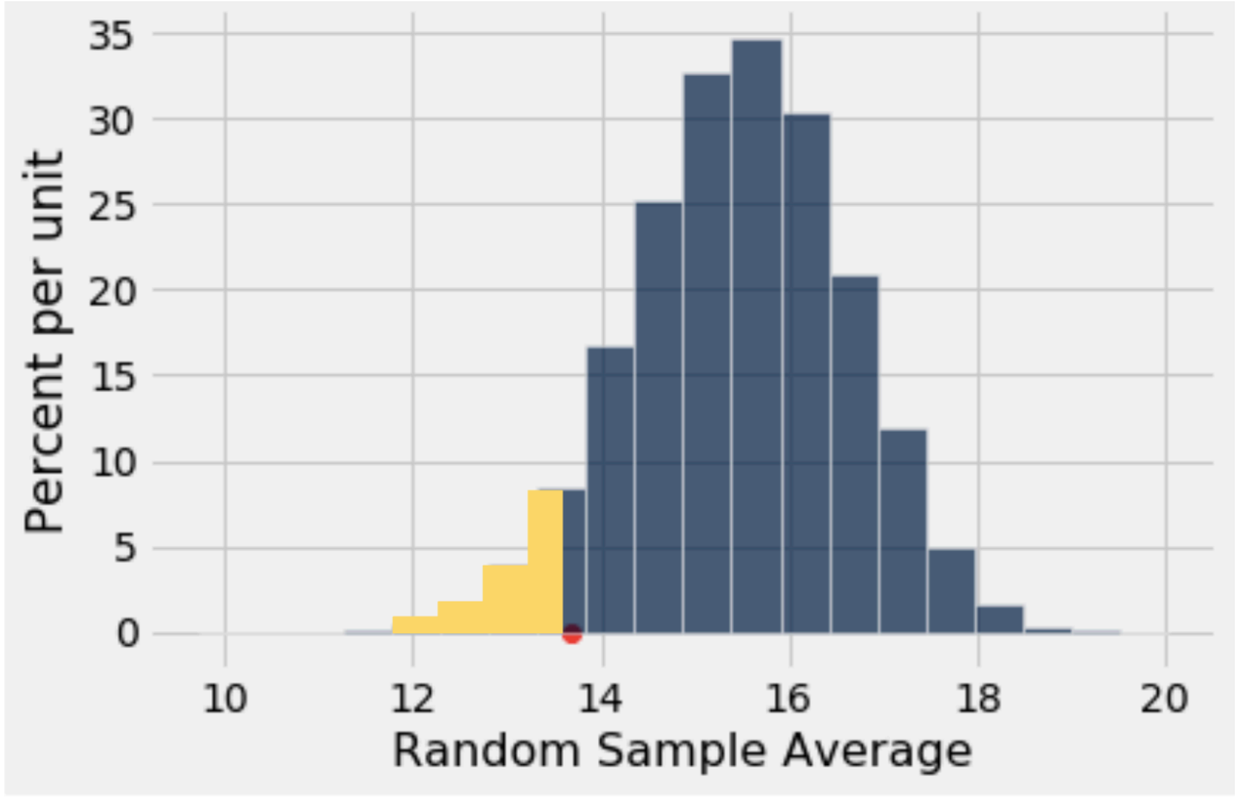
Discussion Questions

What does the red dot represent?

A. Our p-value

B. Our expected outcome

C. Our observed outcome



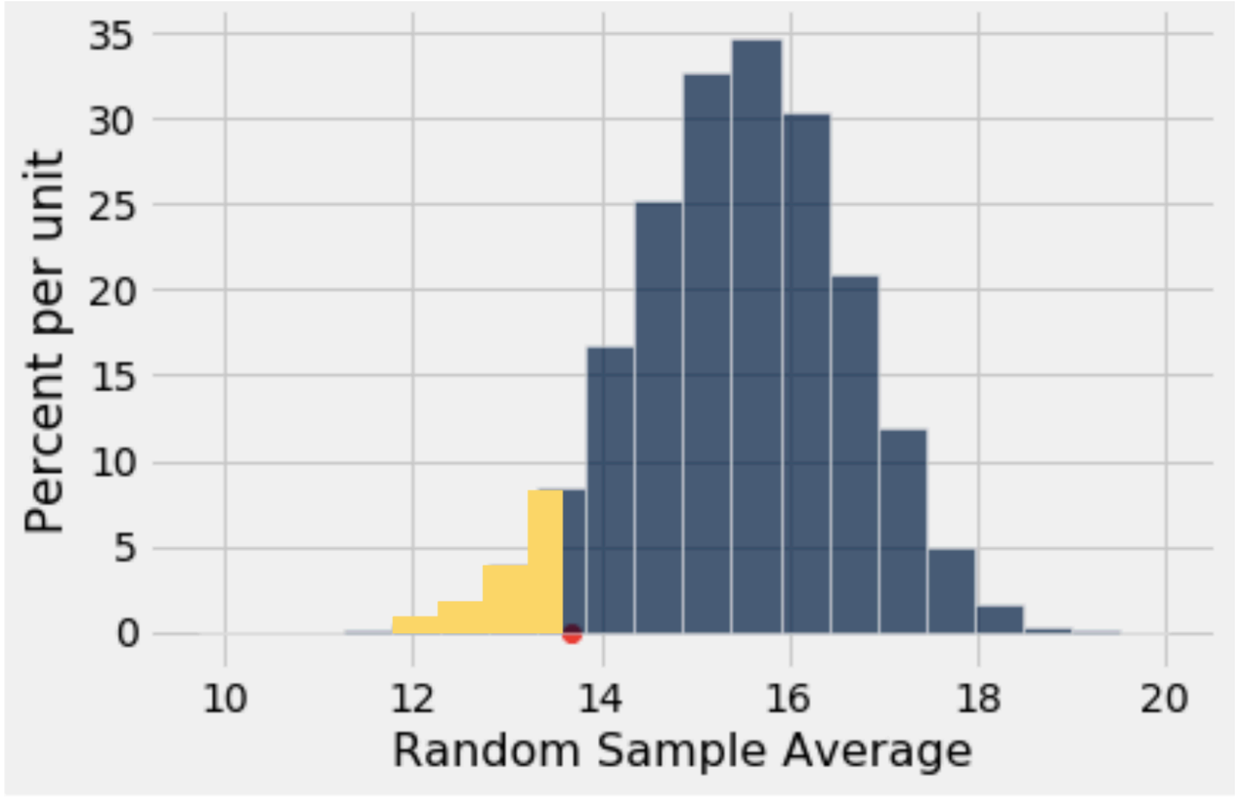
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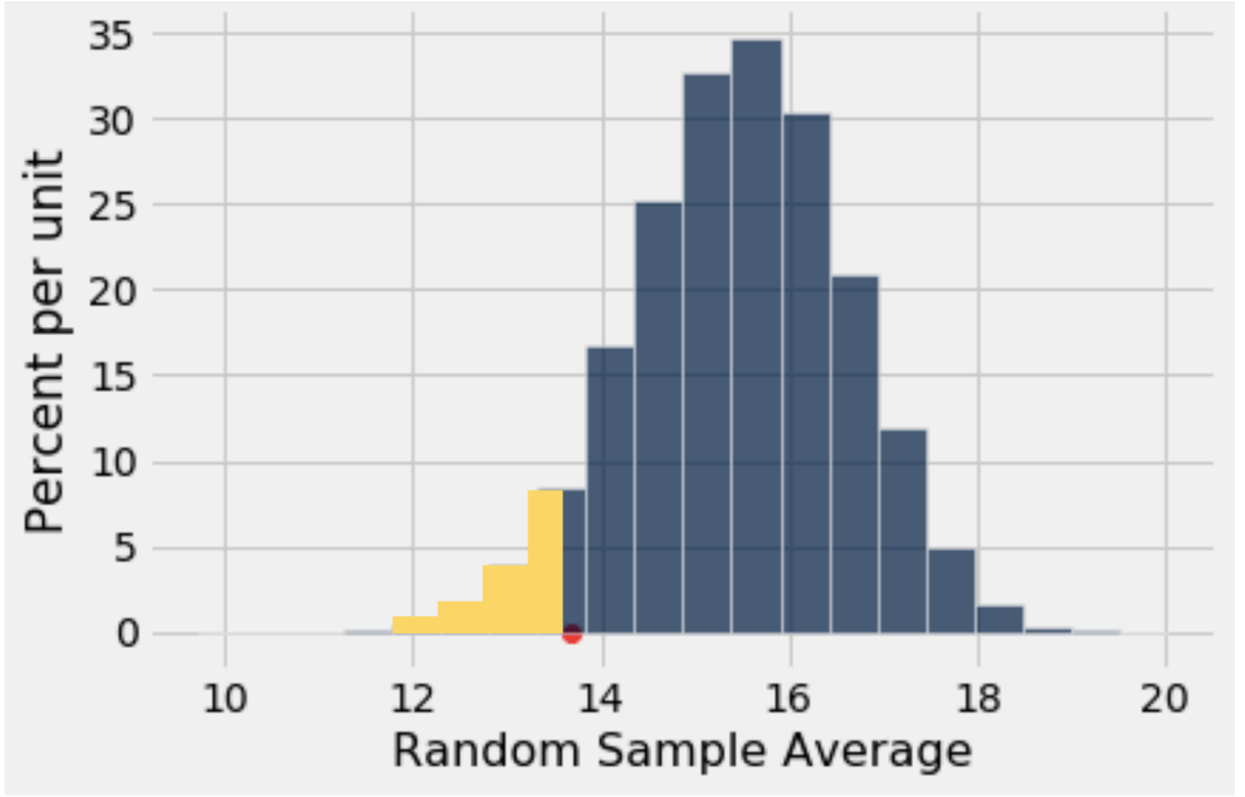
Discussion Questions

What do the yellow bars in this figure represent?

A. The tail of the distribution

B. Our level of statistical significance

C. The probability of getting our observed outcome



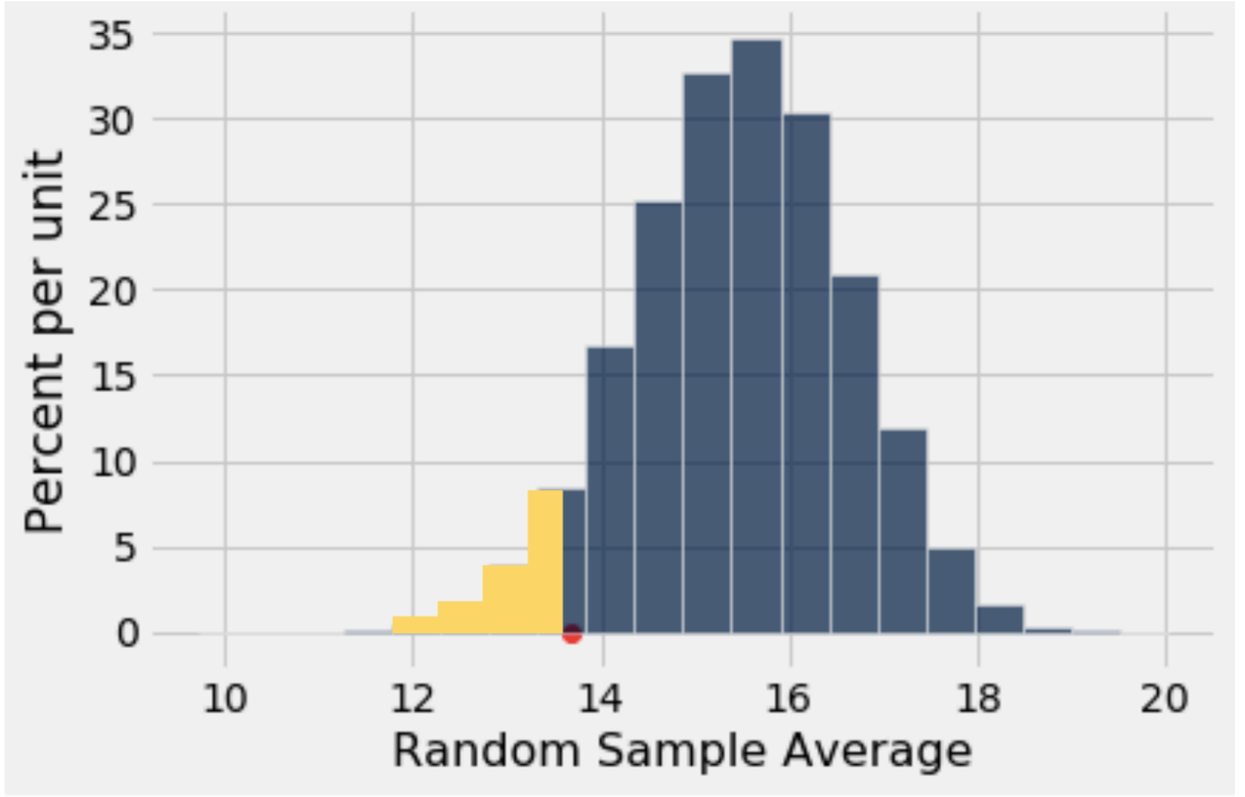
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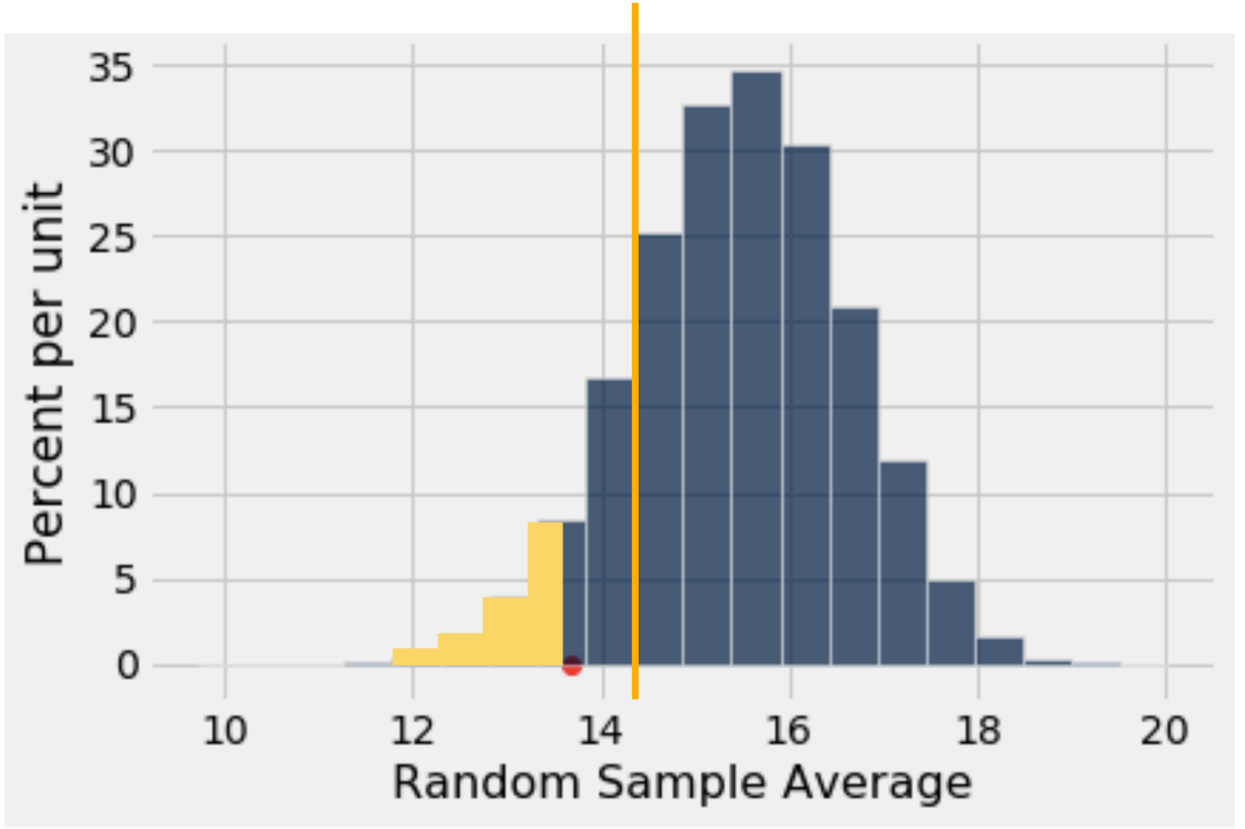
Discussion Questions

Imagine that the yellow vertical bar at ~15 represents our 5% threshold

Which of the following are true:

A. We can reject the null hypothesis, and our result is statistically significant at a 5% threshold

B. We cannot reject the null hypothesis



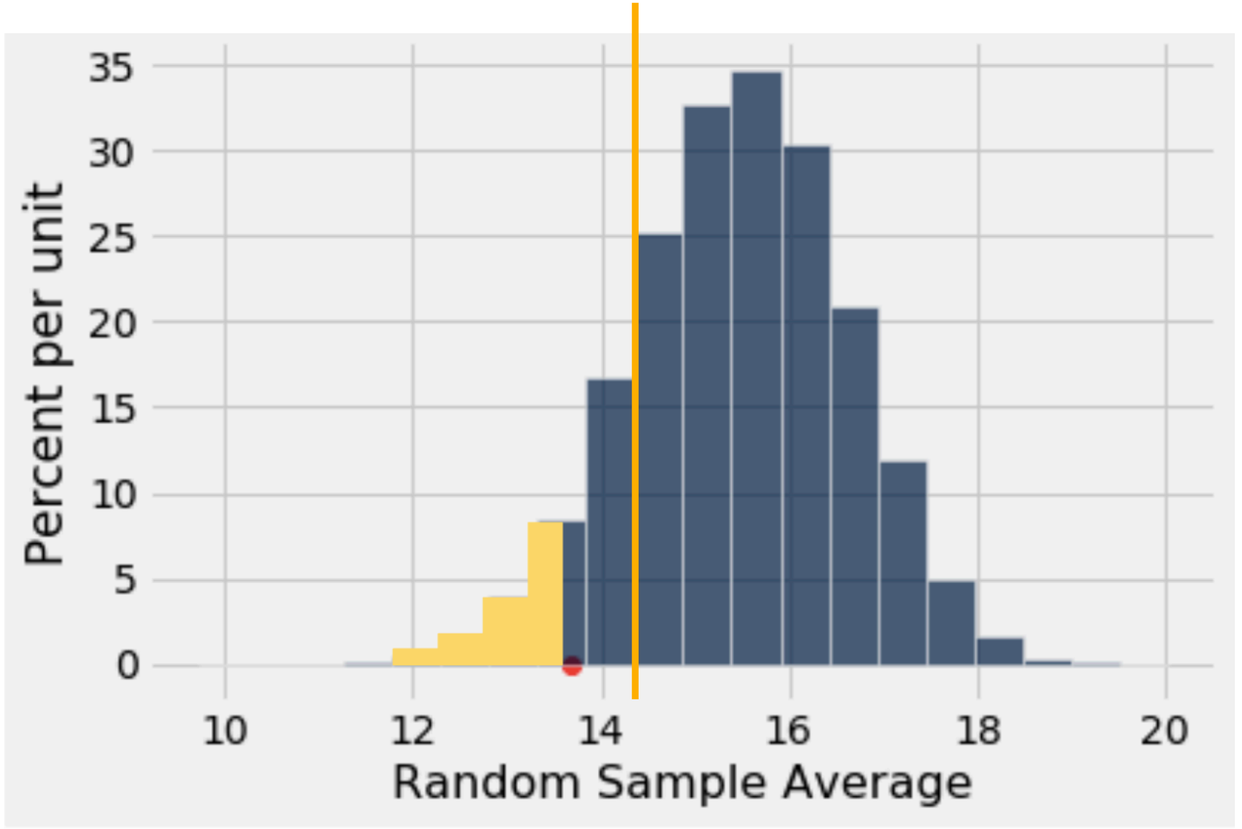
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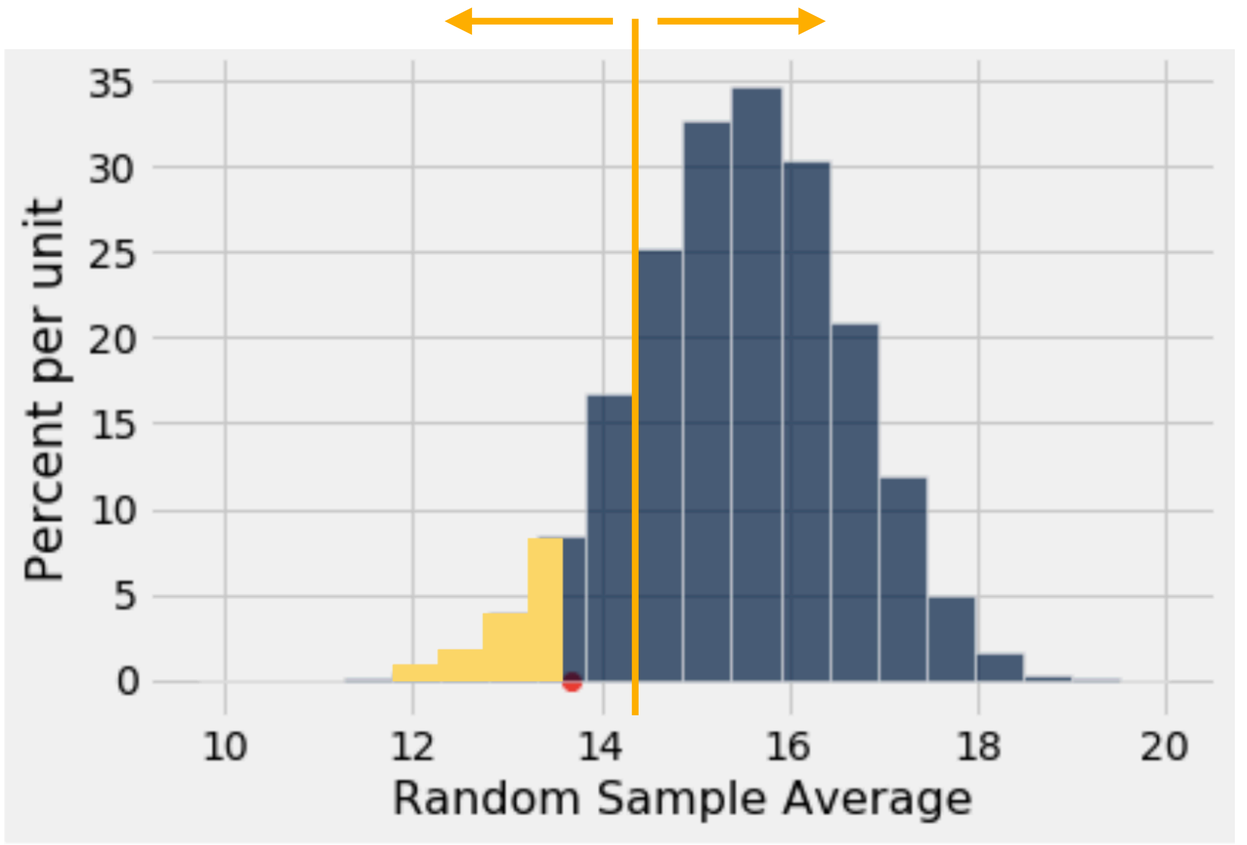
Discussion Questions

Imagine that the yellow vertical bar at ~15 represents our 5% threshold.

Do we expect the 1% threshold to lie:

A. To the left of 15

B. To the right of 15



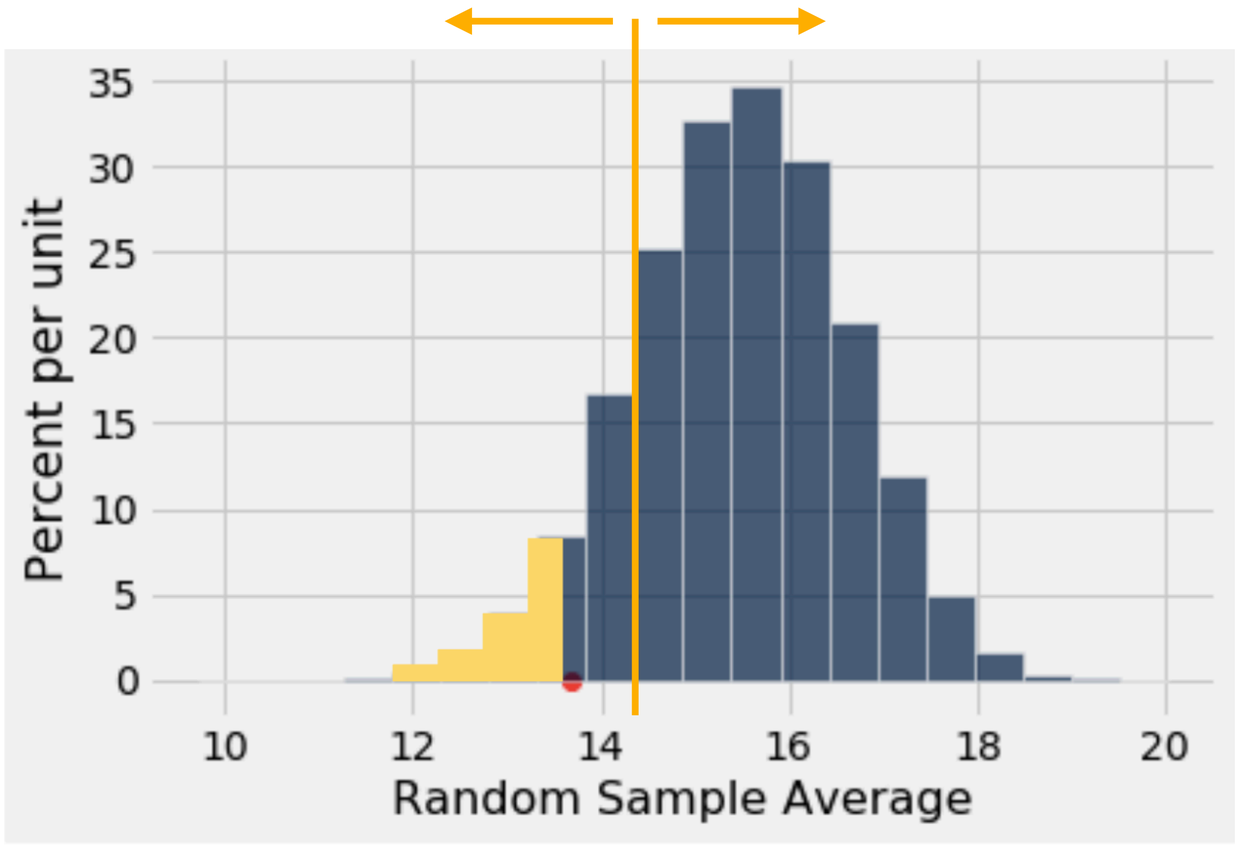
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Summary of Hypothesis Testing so Far

Hypothesis Testing Review

One Category *(e.g. percent of flowers that are purple)*

- Test Statistic (1): `observed_proportion`
- Test Statistic (2): `abs(observed_proportion - null_proportion)`
- Simulate with: `sample_proportions(n, null_dist)`

Multiple Categories *(e.g. ethnicity distribution of jury panel)*

- Test Statistic: `tvd(observed_distribution, null_distribution)`
- Simulate with: `sample_proportions(n, null_distribution)`

Numerical Data *(e.g. scores in a lab section)*

- Test Statistic: `observed_mean`
- Simulate with: `population_data.sample(n, with_replacement=False)`

Next Class

- Today
  - Hypothesis Testing
  - Statistical Significance
- Monday
  - AB Testing